MATH2241
Introduction to Mathematical Analysis
Review Session 1

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Overview

1. Basic Information
2. Chapter 1: The Real Number System
3. Chapter 2: Sequences and Series
4. Practice Questions
5. Summary

Basic Information

Information about me:

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- Feel free to contact me to ask questions

Information about the review session:

- This is the first review session for MATH2241, covering the material for **Test 1**.
- Scope:
  - **Chapter 1**: The Real Number System
  - **Chapter 2**: Sequences and Series
- We will go through some Key Concepts, then work through some Practice Questions.

Key Concepts: Fields and Ordered Fields

Definition (Field)

A field F is a set with $+$ and $\cdot$ satisfying commutativity, associativity, existence of identities $0, 1$, existence of negatives and (nonzero) multiplicative inverses, and distributivity.

Definition (Ordered field)

F is ordered if there is a relation $\leq$ satisfying:

- (O1) For any $a, b \in F$, $a \leq b$ or $b \leq a$.
- (O2) If $a \leq b$ and $b \leq a$, then $a = b$.
- (O3) If $a \leq b$ and $b \leq c$, then $a \leq c$.
- (O4) If $a \leq b$, then $a + c \leq b + c$.
- (O5) If $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

Key Concepts: Bounds, Maximum and Minimum

Let S be a non-empty subset of an ordered field F.

Definition

S is bounded above if there is $M \in F$ with $x \leq M$ for all $x \in S$, and bounded below if there is $m \in F$ with $m \leq x$ for all $x \in S$.

Definition

The maximum of S is $s_0 \in S$ with $x \leq s_0$ for all $x \in S$; the minimum is $i_0 \in S$ with $i_0 \leq x$ for all $x \in S$.

Remarks

- Even if S is bounded above, the maximum may not exist.
- If a maximum/minimum exists, it is unique.

Key Concepts: Supremum and Infimum

Definition (Supremum / Infimum)

If S is bounded above, $\sup S$ is the least upper bound of S. If S is bounded below, $\inf S$ is the greatest lower bound of S.

Remarks

- $\sup S$ or $\inf S$ need not belong to S; if they exist, they are unique.
- If $\max S$ exists, $\max S = \sup S$; similarly for $\min S = \inf S$.

Theorem (Criterion for supremum)

An upper bound u of S equals $\sup S$ ***if and only if*** *for all $\epsilon > 0$, there exists $x \in S$ with $u - \epsilon < x$.*

Key Concepts: Completeness Axiom

Definition

An ordered field F is complete if every non-empty bounded-above subset of F has a supremum in F. $\mathbb{R}$ is defined to be the complete ordered field.

Key facts

- $\mathbb{Q}$ is not complete (for example, $\{x \in \mathbb{Q} : x^2 < 2\}$ has no supremum in $\mathbb{Q}$).
- Completeness also gives: every non-empty bounded-below subset of $\mathbb{R}$ has an infimum (apply completeness to $-S$).

Key Corollary Chain: Archimedean Property and Denseness

1. $\mathbb{N}$ is not bounded above in $\mathbb{R}$.
2. For any $x > 0$, there exists $n_x \in \mathbb{N}$ with $n_x - 1 \leq x < n_x$.
3. For any $x > 0$, there exists $n \in \mathbb{N}$ with $\dfrac{1}{n} < x$.
4. **(Denseness of $\mathbb{Q}$)** If $a < b$ in $\mathbb{R}$, there exists $r \in \mathbb{Q}$ with $a < r < b$.

The set of irrationals is also dense in $\mathbb{R}$.

Key Conclusion: Uncountability of $\mathbb{R}$

Theorem (Nested Interval Property)

If $I_1 \supseteq I_2 \supseteq \cdots$ are non-empty closed bounded intervals, then $\bigcap_n I_n \neq \emptyset$.

Consequence

- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are all countably infinite, but $\mathbb{R}$ is uncountable.
- Since $\mathbb{Q}$ is countable and $\mathbb{R}$ is not, the set of irrational numbers must be uncountable.

Key Concepts: Convergence (ϵ–N Definition)

Definition

We say that a sequence (x_n) converges to x, written as $x_n \to x$ or $\lim x_n = x$, if for every $\epsilon > 0$ there exists N such that $|x_n - x| < \epsilon$ for all $n > N$.

Theorem (Uniqueness of limit)

A convergent sequence has a unique limit.

Theorem

Any convergent sequence in $\mathbb{R}$ is bounded.

- *The converse is false.*
- *For example, $((-1)^n)$ is bounded but not convergent.*

Key Concepts: Algebra of Limits

Suppose $\lim x_n = x$ and $\lim y_n = y$.

- For any $c \in \mathbb{R}$, the sequence (cx_n) converges to cx.
- The sequence $(x_n + y_n)$ converges to $x + y$, and the sequence $(x_n - y_n)$ converges to $x - y$.
- The sequence $(x_n y_n)$ converges to xy.
- If $x_n \neq 0$ for all n and $x \neq 0$, then the sequence $(1/x_n)$ converges to $1/x$.

Theorem (Squeeze Theorem)

If $x_n \leq y_n \leq z_n$ *and* $\lim x_n = \lim z_n = w$, *then* (y_n) *converges and* $\lim y_n = w$.

Key Concepts: Monotone Convergence Theorem

Definition

A sequence (x_n) is increasing if $x_1 \leq x_2 \leq \cdots$.
It is decreasing if $x_1 \geq x_2 \geq \cdots$.
It is monotone if it is either increasing or decreasing.

Theorem (Monotone Convergence Theorem)

*A monotone sequence converges **if and only if** it is bounded.*

Corollary

If (x_n) is bounded and increasing, then $\lim x_n = \sup\{x_n\}$. If (x_n) is bounded and decreasing, then $\lim x_n = \inf\{x_n\}$.

Key Concepts: Subsequences

Definition

A subsequence of (x_n) is (x_{n_k}) for a strictly increasing $n_1 < n_2 < \cdots$ in $\mathbb{N}$.

Theorem

If $\lim x_n = x$, then every subsequence of (x_n) also converges to x.

Divergence criterion

If (x_n) has two subsequences converging to *different* limits, then (x_n) diverges.

Theorem (Monotone Subsequence Theorem)

Every sequence in $\mathbb{R}$ has a monotone subsequence.

Key Concepts: Bolzano–Weierstrass Theorem

Theorem (Bolzano–Weierstrass)

Every bounded sequence of real numbers has a convergent subsequence.

Idea of proof

By the Monotone Subsequence Theorem, the sequence has a monotone subsequence. This subsequence is bounded, so by the Monotone Convergence Theorem, it converges.

Key Concepts: Cauchy Sequences

Definition

(x_n) is Cauchy if for every $\epsilon > 0$, there exists N such that $|x_n - x_m| < \epsilon$ for all $n, m > N$.

Key theorems

- Every convergent sequence is Cauchy.
- Every Cauchy sequence is bounded.
- **(Completeness of $\mathbb{R}$)** A sequence in $\mathbb{R}$ is Cauchy **if and only if** it is convergent.

Key Concepts: Series and Basic Tests

Definition

For a sequence (x_n), the series $\sum x_k$ converges if $\lim_{k\to\infty} s_k$ exists, where $s_k = \sum_{n=1}^{k} x_n$.

Key facts

- Geometric series $\sum r^k$ converges if and only if $0 < r < 1$ (for $r > 0$), to $\frac{r}{1-r}$.
- Harmonic series $\sum \frac{1}{n}$ diverges; p-series $\sum \frac{1}{n^p}$ converges for $p \geq 2$.
- **(n-th term test)** If $\sum x_n$ converges, then $\lim x_n = 0$ (contrapositive tests divergence).
- For non-negative (x_n): $\sum x_n$ converges **if and only if** partial sums are bounded (via the Monotone Convergence Theorem).

Key Concepts: Comparison Tests

Theorem (Comparison Test)

Suppose $0 \leq x_n \leq y_n$ *for all* n. *If* $\sum y_n$ *converges, then* $\sum x_n$ *converges; if* $\sum x_n$ *diverges, then* $\sum y_n$ *diverges.*

Theorem (Limit Comparison Test)

Let $x_n, y_n > 0$ *and* $r = \lim(x_n/y_n)$.

- *If* $r \neq 0$: $\sum x_n$ *converges* ***if and only if*** $\sum y_n$ *converges.*
- *If* $r = 0$ *and* $\sum y_n$ *converges, then* $\sum x_n$ *converges.*

Note that when $r = 0$ *and* $\sum x_n$ *converges,* $\sum y_n$ *need not converge. (Counterexample:* $x_n = 1/n^2$, $y_n = 1/n$*)*

Key Concepts: Absolute and Conditional Convergence

Definition

$\sum x_k$ is absolutely convergent if $\sum |x_k|$ converges; it is conditionally convergent if it converges but not absolutely.

Theorem

Every absolutely convergent series is convergent.

Theorem (Alternating Series Test)

If (x_n) is decreasing, strictly positive, and $\lim x_n = 0$, then the alternating series $\sum (-1)^{n+1} x_n$ converges.

Rearrangement Theorem

Rearranging an absolutely convergent series does not change its sum.

- This can fail for conditionally convergent series.

Key Concepts: Root and Ratio Tests

Theorem (Root Test)

Let $r = \lim \sqrt[n]{|x_n|}$.

- *If* $r < 1$, *then* $\sum x_n$ *is absolutely convergent.*
- *If* $r > 1$, *then* $\sum x_n$ *diverges.*
- *If* $r = 1$, *the test is inconclusive.*

Theorem (Ratio Test)

Let $r = \lim \left| \frac{x_{n+1}}{x_n} \right|$.

- *If* $r < 1$, *then* $\sum x_n$ *is absolutely convergent.*
- *If* $r > 1$, *then* $\sum x_n$ *diverges.*
- *If* $r = 1$, *the test is inconclusive.*

Practice Question 1

Question

Let A be a non-empty subset of $\mathbb{R}$ and let α be an upper bound of A. Suppose that for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $\alpha - \frac{1}{n} < a_n$. Prove that $\alpha = \sup A$.

Suggested Solution 1

Answer

Since α is an upper bound of A, we must have $\alpha \geq \sup A$.
Assume for contradiction that $\alpha > \sup A$. Let $r := \alpha - \sup A > 0$.
Choose $n \in \mathbb{N}$ such that $n > \frac{1}{r}$. Then $r > \frac{1}{n}$ and thus there exists $a_n \in A$ such that

$$\sup A = \alpha - r < \alpha - \frac{1}{n} < a_n.$$

This contradicts that $\sup A$ is an upper bound of A. Therefore, $\alpha = \sup A$.

Practice Question 2

Question

Give an example of each of the following, or prove that such a situation is impossible:

1. $\{a_n\}$ converges, $\{b_n\}$ unbounded, $\{a_n b_n\}$ converges.
2. $\{a_n\}$ unbounded, $\{b_n\}$ converges, $\{a_n - b_n\}$ bounded.
3. $\{a_n\}$ converges, $\{b_n\}$ diverges, $\{a_n + b_n\}$ converges.
4. $\{a_n\}, \{b_n\}$ both diverge, but $\{a_n + b_n\}$, $\{a_n b_n\}$, $\{a_n / b_n\}$ all converge.

Suggested Solution 2 (1/2)

Answer

1. Take $\{a_n\} = \{0, 0, 0, \ldots\}$, $\{b_n\} = \{1, 2, 3, \ldots\}$. Then $\{a_n\} \to 0$, $\{b_n\}$ is unbounded, and $\{a_n b_n\} \to 0$.

2. *Impossible.* Since $\{b_n\}$ converges, it is bounded: $|b_n| \leq M_1$. Since $\{a_n - b_n\}$ is bounded, $|a_n - b_n| \leq M_2$. As $a_n = (a_n - b_n) + b_n$,

$$|a_n| \leq |a_n - b_n| + |b_n| \leq M_1 + M_2,$$

so $\{a_n\}$ must also be bounded.

Suggested Solution 2 (2/2)

Answer

3. *Impossible.* If $\{a_n + b_n\}$ and $\{a_n\}$ converge, then $\{b_n\} = \{(a_n + b_n) - a_n\}$ also converges.

4. Take $\{a_n\} = \{1, -1, 1, -1, \ldots\}$, $\{b_n\} = \{-1, 1, -1, 1, \ldots\}$. Both diverge, but

$$\begin{aligned} \{a_n + b_n\} &= \{0, 0, 0, \ldots\}, \\ \{a_n b_n\} &= \{-1, -1, -1, \ldots\}, \\ \{a_n / b_n\} &= \{-1, -1, -1, \ldots\} \end{aligned}$$

are all convergent.

Practice Question 3

Question

Let $a > 1$. Define a sequence $\{x_n\}$ by

$$x_1 = a, \qquad x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) \quad \text{for any } n \in \mathbb{N}.$$

1. Prove by induction that $x_n > \sqrt{a}$ for any $n \in \mathbb{N}$.
2. Prove that $\{x_n\}$ is decreasing. Hence show that $\{x_n\}$ converges. Find also the limit.

Suggested Solution 3 (1/2)

Answer

1. We have $x_1 = a > \sqrt{a}$ since $a > 1$. Assume $x_k > \sqrt{a}$; then $(x_k - \sqrt{a})^2 > 0$, so

$$x_k^2 - 2x_k\sqrt{a} + a > 0 \implies x_k^2 + a > 2x_k\sqrt{a}$$
$$\implies \frac{x_k^2 + a}{2x_k} > \sqrt{a} \implies x_{k+1} > \sqrt{a}.$$

By induction, $x_n > \sqrt{a}$ for all $n \in \mathbb{N}$.

Suggested Solution 3 (2/2)

Answer

2. By (1), $x_n^2 > a$, so $2x_n^2 > x_n^2 + a$, giving

$$x_n > \frac{x_n^2 + a}{2x_n} = x_{n+1}.$$

So $\{x_n\}$ is decreasing and bounded below by $\sqrt{a}$; by the Monotone Convergence Theorem, $\{x_n\}$ converges. If $L = \lim x_n$, then

$$L = \frac{L^2 + a}{2L} \implies L^2 = a.$$

Since all terms are positive, $L = \sqrt{a}$.

Practice Question 4

Question

Let $x_1 := a > 0$ and $x_{n+1} := x_n + 1/x_n$ for $n \in \mathbb{N}$. Determine whether (x_n) converges or diverges.

Suggested Solution 4

Answer

Assume (x_n) converges. Then (x_n) is bounded: $|x_n| \leq u$ for some $u \in \mathbb{R}$. Since $x_1 = a > 0$, $u > 0$. Then $|x_{n+1} - x_n| = \left|\frac{1}{x_n}\right| \geq \left|\frac{1}{u}\right|$ for all n.

Taking $\epsilon = \left|\frac{1}{u}\right| > 0$: for any $N \in \mathbb{N}$, taking $m = N + 1$, $n = N$ gives $|x_m - x_n| \geq \epsilon$, so (x_n) is **not** Cauchy, hence does **not** converge, giving a contradiction. Therefore, (x_n) diverges.

Practice Question 5

Question

Let $0 < \alpha < 1$ and let $\{a_n\}$ satisfy

$$|a_{n+2} - a_{n+1}| \leq \alpha |a_{n+1} - a_n| \qquad \text{for all } n \in \mathbb{N}.$$

Prove that $\{a_n\}$ is a Cauchy sequence (and hence converges).

Suggested Solution 5

Answer

The condition gives $|a_{n+2} - a_{n+1}| \leq \alpha^n |a_2 - a_1|$. If $a_1 = a_2$, $\{a_n\}$ is constant, hence Cauchy. Assume $a_1 \neq a_2$. For $n > m \geq 2$,

$$\begin{aligned} |a_n - a_m| &\leq |a_n - a_{n-1}| + \cdots + |a_{m+1} - a_m| \\ &\leq (\alpha^{n-2} + \cdots + \alpha^{m-1})|a_2 - a_1| \\ &< \alpha^{m-1} \left(\frac{1}{1-\alpha} \right) |a_2 - a_1| = C\alpha^{m-1}. \end{aligned}$$

Since $0 < \alpha < 1$, $\alpha^m \to 0$. So for any $\epsilon > 0$, there is $N \in \mathbb{N}$ such that $m \geq N$ implies $\alpha^{m-1} < \dfrac{1}{C}\epsilon$.

It follows that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n > m \geq N$ implies $|a_n - a_m| < \epsilon$. This shows $\{a_n\}$ is Cauchy, and hence converges.

Practice Question 6

Question

Determine whether each series converges. Justify your answers.

1. $\sum \dfrac{n^2}{n!}$
2. $\sum \left[\sqrt{n+1} - \sqrt{n}\right]$
3. $\sum \dfrac{n!}{n^n}$
4. $\displaystyle\sum_{n=2}^{\infty} \dfrac{1}{\sqrt{n}\log n}$

Suggested Solution 6 (1/2)

Answer

1. Let $a_n := \dfrac{n^2}{n!}$. Since $\lim \left| \dfrac{a_{n+1}}{a_n} \right| = \lim \left| \dfrac{n+1}{n^2} \right| = 0 < 1$, the series **converges** by the Ratio Test.

2. Let $s_n := \sum_{i=1}^{n} [\sqrt{i+1} - \sqrt{i}]$. This telescopes: $s_n = \sqrt{n+1} - 1 \to \infty$. So the series **diverges** (even though the terms $\to 0$).

Suggested Solution 6 (2/2)

Answer

3. Let $a_n := \dfrac{n!}{n^n}$. Simplifying,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1}\right)^n \longrightarrow e^{-1} < 1,$$

so the series **converges** by the Ratio Test.

4. For large n, $1 \leq \log n \leq \sqrt{n}$, so $\dfrac{1}{\sqrt{n}\log n} \geq \dfrac{1}{n}$. Since $\sum \frac{1}{n}$ diverges, the series **diverges** by the Comparison Test.

Practice Question 7

Question

1. Suppose $(s_n = \sum_{k=1}^{n} a_k)$ is absolutely convergent. Prove that $(t_n = \sum_{k=1}^{n} a_k^3)$ is convergent.
2. More generally, what can be said about $\sum a_n^k$ for other integers k?

Suggested Solution 7 (1/2)

Answer

1. Since (s_n) is absolutely convergent, by the Cauchy criterion, given $\epsilon^{1/3} > 0$ there is N_ϵ with

$$|a_m| + \cdots + |a_{n+1}| < \epsilon^{1/3} \quad \text{for all } m > n \geq N_\epsilon.$$

Then

$$|a_m|^3 + \cdots + |a_{n+1}|^3 \leq (|a_m| + \cdots + |a_{n+1}|)^3 < \epsilon,$$

so (t_n) is absolutely convergent by the Cauchy criterion, hence convergent.

Suggested Solution 7 (2/2)

Answer

2. The same proof works for any integer $k \geq 1$, using $x_1^k + \cdots + x_r^k \leq (x_1 + \cdots + x_r)^k$ for $x_i \geq 0$. So $\sum a_n^k$ **converges (absolutely) for every integer** $k \geq 1$.

   For $k \leq 0$ this generally fails: $k = 0$ gives $\sum 1$ (diverges); for $k < 0$, since $a_n \to 0$, we get $a_n^k = 1/a_n^{|k|} \to \infty$ along terms with $a_n \neq 0$, so $\sum a_n^k$ **diverges** by the n-th term test (or is undefined).

Practice Question 8

Question

1. Find a series $\sum a_n$ which diverges by the Root Test but for which the Ratio Test gives no information.
2. Find a series $\sum a_n$ which converges by the Root Test but for which the Ratio Test gives no information.

Suggested Solution 8 (1/2)

Answer

1. Let $a_n = 2^{n+(-1)^n}$.

   Since $\sqrt[n]{a_n} = 2^{1+(-1)^n/n} \to 2 > 1$, the Root Test shows $\sum a_n$ diverges.

   But $\dfrac{a_{n+1}}{a_n} = 2^{1-2(-1)^n}$ alternates forever between 8 (n odd) and $\frac{1}{2}$ (n even), so the limit does not exist and the Ratio Test gives no information.

Suggested Solution 8 (2/2)

Answer

2. Let $a_n = 2^{(-1)^n - n}$.

   Since $\sqrt[n]{a_n} = 2^{(-1)^n/n - 1} \to \frac{1}{2} < 1$, the Root Test shows $\sum a_n$ converges (absolutely).

   But $\dfrac{a_{n+1}}{a_n} = 2^{-2(-1)^n - 1}$ alternates forever between 2 (n odd) and $\frac{1}{8}$ (n even), so again the limit does not exist and the Ratio Test gives no information.

Remark

In general, for a sequence (a_n) with $a_n > 0$ for all n, we have:

$$\liminf \frac{a_{n+1}}{a_n} \leq \liminf (a_n)^{1/n} \leq \limsup (a_n)^{1/n} \leq \limsup \frac{a_{n+1}}{a_n},$$

which is why root test is often considered 'more powerful' than the ratio test.

Key Takeaways

- **Chapter 1**: $\sup/\inf$ need not belong to the set; completeness of $\mathbb{R}$ underlies everything (denseness, existence of irrationals, uncountability).
- **Chapter 2**: Three main convergence tools: the ϵ–N definition, the Monotone Convergence Theorem (monotone and bounded), and the Cauchy criterion; the Bolzano–Weierstrass theorem gives convergent subsequences of bounded sequences; series convergence via comparison, limit comparison, root, and ratio tests; absolute convergence is the strongest and most robust notion.
- **Exam strategy**: Identify which definition or theorem is being tested, and write ϵ–N proofs carefully and completely.

Group Chats

- As instructed by the department, I am also forming group chats to serve as a platform to provide assistance and to encourage collaboration among students.

WhatsApp Group



WeChat Group



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